

Hidden in Plain Sight: Occlusion Edge Blur as a Perceptual Blind Spot in Multimodal LLMs

Jessica Herrington
School of Cybernetics
Australian National University
Canberra, ACT, Australia
jessica.herrington@anu.edu.au

Ben Swift
School of Cybernetics
Australian National University
Canberra, ACT, Australia
ben.swift@anu.edu.au

Abstract

Multimodal large language models (MLLMs) have demonstrated impressive reasoning capabilities across a range of visual tasks, yet emerging evidence suggests they struggle with fundamental perceptual processes. We investigate this gap using occlusion edge blur: a depth cue that humans exploit effortlessly but that cannot be solved by reasoning about a verbal description of the image by MLLMs. Our findings reveal a systematic bias in MLLMs likely driven by a verbal heuristic (“sharp edge means closer”) that overrides actual visual information in an image. That is, MLLMs can be easily ‘fooled’ on depth information by blurring the edge of an occluding object. This has implications for disinformation detection, robustness to adversarial attacks, and the design of new evaluation benchmarks.

CCS Concepts

• **Computing methodologies** → **Computer vision problems**: *Neural networks*; • **Human-centered computing** → *Empirical studies in HCI*.

Keywords

multimodal large language models, visual perception, occlusion edge blur, depth perception, disinformation detection, psychophysics

ACM Reference Format:

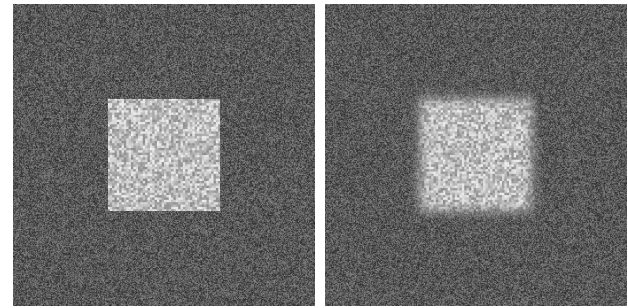
Jessica Herrington and Ben Swift. 2026. Hidden in Plain Sight: Occlusion Edge Blur as a Perceptual Blind Spot in Multimodal LLMs. In *The 5th ACM International Workshop on Multimedia AI against Disinformation (MAD '26)*, June 16–19, 2026, Amsterdam, Netherlands. ACM, New York, NY, USA, 8 pages. <https://doi.org/10.1145/3810988.3812661>

1 Introduction

Multimodal large language models (MLLMs), the vision-capable AI systems behind tools such as ChatGPT and Claude, are increasingly deployed as components in multimedia forensics and disinformation detection pipelines, assessing image authenticity, flagging manipulated content, and serving as visual reasoning backends for fact-checking systems. The reliability of these systems depends on whether MLLMs can accurately perceive both low-level and high-level visual properties, not merely reason about verbal descriptions of images [3, 16, 21].



This work is licensed under a Creative Commons Attribution 4.0 International License. *MAD '26*, Amsterdam, Netherlands
© 2026 Copyright held by the owner/author(s).
ACM ISBN 979-8-4007-2700-9/26/06
<https://doi.org/10.1145/3810988.3812661>



(a) Crisp occlusion edge (b) Blurred occlusion edge

Figure 1: Occlusion edge blur as a depth cue. Both images show a textured square occluding a textured background. In (a) the occlusion edge is crisp and the square appears closer to the viewer. In (b) the occlusion edge is blurred and the square appears further away, even though it is still the occluding surface. Figure inspired by Marshall et al. [12].

Simple phenomena such as occlusion and blur can provide an ideal probe to examine limitations in MLLM perceptual capability. The occlusion edge (where objects overlap) is particularly informative: yet a persistent challenge when viewing natural scenes is that blurry objects could be either very close to or very far from the viewer. That is, blurry objects could occlude or be occluded.

In human vision, *occlusion edge blur* is a depth cue. That is, a blurred surface is perceived as being closer or further away (see Figure 1) depending on whether the occlusion edge belonging to an object is blurred or crisp [12, 13, 17]. Critically, in humans, blur and depth order is reliant on early visual processing in retinal pathways and brain area V1, and is not language based [15]. This means that you cannot reason your way to the correct answer from a verbal description alone. This makes occlusion edge blur a test of whether MLLMs genuinely process visual features or instead fall back on verbal heuristics to reason about visual scenes.

This evidence motivates a specific hypothesis: if MLLMs fall back on verbal heuristics when interpreting blur, they should tend to judge the *blurred objects* as being further away and crisp objects as being closer even when the opposite is true. Moreover, if this bias is driven by a learned association rather than task confusion, it should grow stronger as blur increases, rather than a flat or random pattern.

We present a controlled psychophysics-style experiment testing six MLLMs on an occlusion edge blur depth-order task across varying blur magnitudes, implementing several prompting strategies. In

the task, two circles intersect, one has a crisp edge and the other has a blurred edge (Figure 2). The task requires the MLLM to perceive depth from occlusion cues and determine which circle is ‘in front’ of the other.

2 Related work

2.1 Perceptual deficits in MLLMs

Recent work suggests that vision remains less developed relative to language in current MLLM architectures, with models showing widespread deficits on standardised perceptual tests [1, 8, 20]. Fu et al. [3] show that MLLMs can “see” but not “perceive”, failing on tasks that require low-level visual processing rather than high-level reasoning. Rahmazadehgervi et al. [16] go further, demonstrating failures on visual tasks so elementary that the title of their paper is simply “Vision Language Models Are Blind”. Tong et al. [21] identify systematic visual shortcomings across multiple model families, finding that these deficits persist even as language capabilities improve.

If MLLMs harbour systematic blind spots in how they interpret fundamental visual features, adversaries could exploit these to create manipulations that evade automated detection while remaining obvious to human observers [10].

2.2 Occlusion perception and blur

This relationship between edge blur and perceived depth order provides a robust, low-level perceptual signal [17]. Recent work confirms that occlusion perception remains a considerable challenge for MLLMs, with a significant performance gap compared to humans [11]. Notably, “neither scaling nor ‘thinking’ in MLLMs is a sufficient strategy for mastering occlusion perception” [11], reinforcing the view that these models have significant perceptual rather than reasoning deficits.

Early efforts in this direction, such as illusion benchmarks where chain-of-thought prompting actually *degrades* performance by reinforcing initial perceptual errors [7], reinforce the view that these failures are perceptual rather than reasoning-based.

2.3 Psychophysical evaluation of MLLMs

To better understand the perceptual capability of MLLMs, researchers are increasingly advocating for the use of a psychophysical toolkit—incorporating signal detection, threshold estimation, and parametric stimulus control—to be applied systematically to machine perception, not just human perception [2, 6]. Ghosh et al. [4] propose the Percept-V challenge, a benchmark of simple perception problems designed to test whether MLLMs can crack tasks that humans find trivial. Hernández-Cámara et al. [6] apply contrast sensitivity functions—a cornerstone of human psychophysics—to evaluate MLLM vision, finding systematic deviations from human performance profiles.

Our work builds on this tradition by using occlusion edge blur as a targeted psychophysical probe: a stimulus manipulation grounded in human visual perception research that isolates a specific perceptual process and tests whether MLLMs can perform it.

3 Method

3.1 Stimuli

Each stimulus image depicts two overlapping circles on a 512 × 512 px mid-grey background (RGB 128, 128, 128). Both circles have a radius of 100 px with centres offset by 75 px, producing approximately 25% area overlap. One circle is rendered with crisp edges; the other is Gaussian-blurred. The circles are drawn in distinct colours selected from six OKLCH hues ($L=0.7$, $C=0.15$: red, yellow, green, cyan, blue, magenta), and are identical in terms of saturation and lightness (Figure 2).

We also developed a sweep of blurred stimuli where the blur radius was increased iteratively from 0 px to 20 px (Figure 3).

3.2 Experimental design

We tested six MLLMs spanning two major providers and a range of model scales: Claude Opus 4.6, Claude Sonnet 4.6, and Claude Haiku 4.5 (Anthropic); GPT-5.4, GPT-5.4 Mini, and GPT-5.4 Nano (OpenAI). We conducted two complementary experiments. A preliminary *full factorial* design (120 conditions) crossed depth order (crisp on top vs. blurred on top), spatial position (crisp circle on left vs. right), and colour pair (all 30 ordered pairs of 6 hues) at a fixed blur radius of 20 px. Results showed no significant effects of colour or position, so a reduced *blur sweep* design (96 conditions) varied blur radius across six levels (0, 4, 8, 12, 16, 20 px) with four representative colour pairs (red/cyan, yellow/blue, green/magenta, cyan/red), retaining depth order and spatial position. The zero-blur condition (0 px) serves as a catch trial: with both circles crisp, any bias attributable to blur should vanish, and the task reduces to basic occlusion detection.

To assess whether the crisp-in-front bias could be mitigated through prompt engineering, we tested six prompt variants (Table 1). *Neutral* describes the image and asks which circle is in front. *Minimal* poses the bare question with no framing. *Foreground* uses explicit foreground/background terminology. *Psychophysics* adopts an experimental framing that draws attention to sharpness and blur. *Scripted CoT* prescribes a fixed reasoning script: it instructs the model to first describe each circle’s visual properties, then examine which edge “continues through” the overlap region, then answer accordingly. *Thinking* uses the same text as *neutral* but enables provider-level reasoning tokens (Anthropic uses “extended thinking”, OpenAI uses “reasoning effort”), allowing the model to generate free-form intermediate reasoning before committing to an answer.¹

Each model was evaluated on every condition with three repetitions, for a binary forced-choice response (left or right). Responses were parsed from JSON where possible, with fallback to free-text extraction. For the scripted-CoT and thinking prompt variants, full reasoning traces were captured alongside the forced-choice response, enabling qualitative analysis of the models’ reasoning processes. Across both experiments, we collected 21,875 trials, of which 21,872 yielded parseable responses (3 unparseable, <0.02%). The thinking prompt was evaluated on the blur sweep conditions only for most

¹We use “Scripted CoT” to distinguish our prompt-engineered chain-of-thought from the free-form intermediate reasoning that current frontier models produce when reasoning tokens are enabled. Much of the recent CoT literature conflates these two regimes; we report them as separate conditions because they differ in both who controls the reasoning trajectory (the prompt author vs. the model) and in how the resulting tokens are weighted by the model.

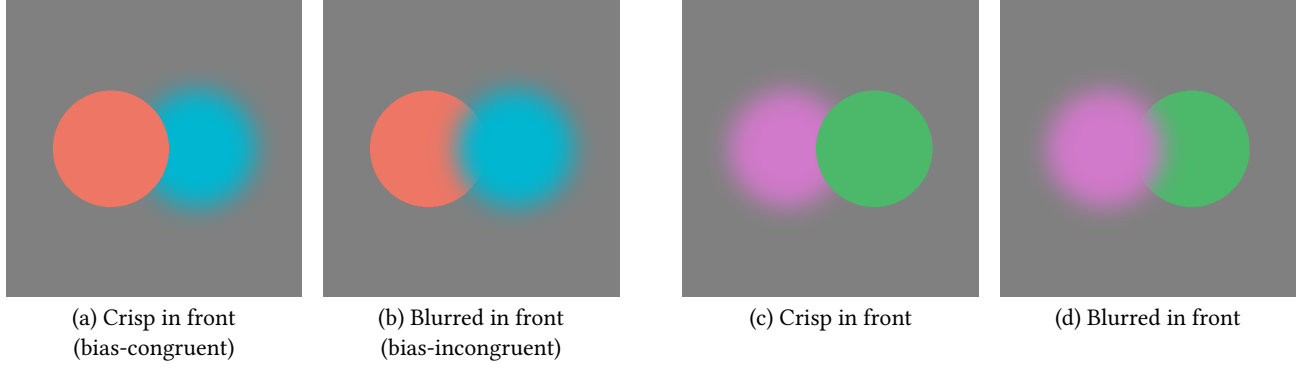


Figure 2: Stimulus examples at 20 px blur. (a, b) Red and cyan circles: (a) the crisp circle occludes the blurred circle (bias-congruent), (b) the blurred circle occludes the crisp circle (bias-incongruent). (c, d) A second colour pair showing the same manipulation. The physical overlap is identical across all images; only the depth order and corresponding blur boundary differ.

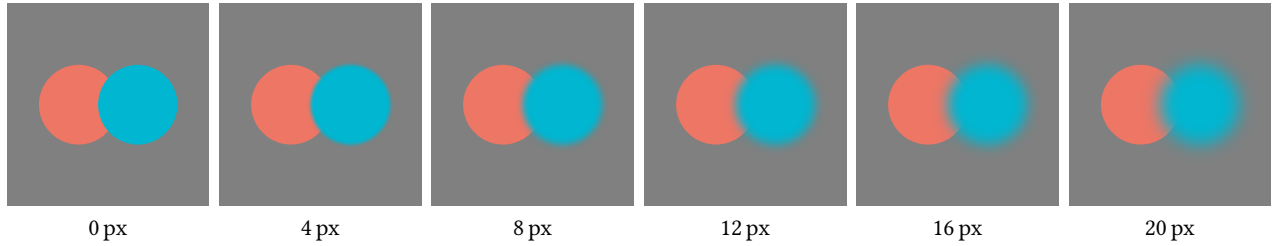


Figure 3: Blur sweep stimuli. Each image shows two overlapping circles: in each case the cyan circle occludes the red circle. For the cyan circle, Gaussian blur radius increases from left (0 px, no blur) to right (20 px). At increasing amounts of blur, the blurred circle’s edge becomes more diffuse, but it remains the occluding surface in all examples pictured.

Table 1: Prompt text for each variant. All prompts elicit a two-alternative forced-choice (2AFC) left/right response.

Variant	Prompt text
Neutral	This image shows two overlapping circles on a grey background. One circle is on the left and one is on the right. One circle is in front of (occluding) the other. Which circle is in front—the left circle or the right circle? Reply with a single JSON object: {"answer": "left"} or {"answer": "right"}.
Minimal	Which circle is in front, the left or the right? Reply with just "left" or "right".
Foreground	Two circles overlap in this image. One is in the foreground and one is in the background. Is the foreground circle on the left side or the right side? Reply with a single JSON object: {"answer": "left"} or {"answer": "right"}.
Psychophysics	You are participating in a visual perception experiment. This image contains two overlapping circles that differ in sharpness—one has crisp edges and the other has blurred edges. Based only on the visual information in the image, which circle appears to be closer to the viewer, the left or the right? Reply with a single JSON object: {"answer": "left"} or {"answer": "right"}.
Scripted CoT	This image shows two overlapping circles on a grey background. One circle is on the left and one is on the right. One circle is in front of (occluding) the other. Before answering, reason step by step: (1) Describe the visual properties of each circle. (2) Look at the overlap region—which circle’s edge continues through, and which is interrupted? (3) The circle whose edge continues uninterrupted is in front. After your reasoning, give your final answer as a JSON object: {"answer": "left"} or {"answer": "right"}.
Thinking	Same text as <i>neutral</i> ; provider-level reasoning tokens enabled (free-form intermediate reasoning, not scripted).

models, as the full factorial was not re-run after reasoning trace capture was corrected. Cross-blur analyses use the balanced blur-sweep subset (10,356 trials) to avoid over-representing the 20 px condition from the full factorial.

4 Results

Each model shows a strong bias toward judging the crisp circle as in front, regardless of which circle actually occludes the other. When the crisp circle is on top (bias-congruent), accuracy is near ceiling; when the blurred circle is on top (bias-incongruent), accuracy drops sharply. Pooled across all models under the five non-thinking prompts, accuracy is 93.6% for bias-congruent trials versus 22.7%

Table 2: Per-model summary under the five non-thinking prompts (*neutral, minimal, foreground, psychophysics, scripted CoT*). Crisp↑ and Blurred↑: accuracy when the crisp or blurred circle is on top. OR: depth-order odds ratio. 0 px: accuracy at the zero-blur catch trial (pooled across both depth orders). Anthropic models are above the rule; OpenAI below. Thinking-prompt results in Table 3.

Model	Crisp↑	Blurred↑	OR	0 px
Haiku 4.5	87.9%	19.1%	30.8	57.1%
Opus 4.6	98.4%	10.7%	500	73.3%
Sonnet 4.6	99.3%	10.3%	1 244	82.5%
GPT-5.4	99.3%	20.2%	600	93.3%
GPT-5.4 Mini	98.0%	35.5%	89.9	82.9%
GPT-5.4 Nano	78.5%	39.9%	5.5	57.1%

Table 3: Per-model summary under the *thinking* prompt (provider-level reasoning tokens enabled, free-form intermediate reasoning). Anthropic models (above the rule) remain strongly biased; OpenAI models (below) show order-of-magnitude reductions in OR. Discussed further in Section 5.

Model	Crisp↑	Blurred↑	OR
Haiku 4.5	93.2%	11.1%	108
Opus 4.6	97.8%	23.6%	143
Sonnet 4.6	98.5%	8.6%	655
GPT-5.4	97.2%	78.5%	9.5
GPT-5.4 Mini	97.9%	72.2%	17.9
GPT-5.4 Nano	89.6%	48.6%	9.0

for bias-incongruent trials. We report the thinking prompt separately (Table 3) because, as we show below, its effect on the bias diverges sharply between providers and would otherwise distort the headline.

We summarise this asymmetry with the *depth-order odds ratio* (OR): the ratio of the odds of a correct response when the crisp circle is on top to the odds of a correct response when the blurred circle is on top,

$$\text{OR} = \frac{\Pr(\text{correct} \mid \text{crisp on top}) / \Pr(\text{wrong} \mid \text{crisp on top})}{\Pr(\text{correct} \mid \text{blur on top}) / \Pr(\text{wrong} \mid \text{blur on top})}.$$

OR = 1 indicates accuracy is independent of depth order (no bias); OR $\gg$ 1 indicates a strong crisp-in-front preference (the model is much more often correct when the crisp circle occludes). We estimate the OR by conditional maximum likelihood; 95% confidence intervals are reported in the analysis pipeline. Per-model depth-order odds ratios under the non-thinking prompts range from 5.5 (GPT-5.4 Nano) to 1 244 (Claude Sonnet 4.6), confirming that every model and provider exhibits the bias, though its magnitude varies considerably (Table 2).

This bias scales monotonically with blur magnitude (Figure 4). Pooled across the non-thinking prompts, accuracy on bias-incongruent trials falls from 72.5% at 0 px blur to 10.8% at 20 px, while the depth-order odds ratio escalates from 1.2 to 209 over the same range.

The 0 px catch trial—where both circles are crisp and the task reduces to plain occlusion detection—is informative in two ways. First, the bias is essentially absent at 0 px: pooled accuracy is 74.4% with OR = 1.2 (95% CI: 1.0–1.6), indistinguishable from no bias, and per-model deviations from convergence are not consistent in direction. The crisp-in-front bias requires an actual blur differential rather than reflecting a baseline response tendency or rendering artefact. Second, baseline competence at simple occlusion detection is heterogeneous: four models cluster between 73% and 93% accuracy at 0 px (Opus 4.6: 73.3%, Sonnet 4.6: 82.5%, GPT-5.4: 93.3%, GPT-5.4 Mini: 82.9%), but two perform near chance (Haiku 4.5: 57.1%, GPT-5.4 Nano: 57.1%). For the smaller models, the stimulus-response curve is best read as the incremental tilting of already-marginal performance rather than the corruption of high-fidelity perception.

Prompt wording has a measurable but much smaller effect than depth order (Figure 5). Accuracy on blurred-on-top trials ranges from 14.3% (*psychophysics*) to 31.5% (*thinking*), and no prompt variant overcomes the bias when pooled across models (Figure 5). Our scripted-CoT prompt is ineffective (20.6%): instructing the model to step through edge continuity does not help, consistent with broader evidence that prompt-engineered CoT primarily benefits symbolic and mathematical tasks [19]. Free-form provider-level reasoning (*thinking*) tells a more interesting story: pooled accuracy on blurred-on-top trials rises to 31.5%, but this average masks large differences between models. GPT-5.4 reaches 78.5% and GPT-5.4 Mini 72.2% with thinking enabled—well above chance—while the Anthropic models show little improvement (Opus 23.6%, Sonnet 8.6%, Haiku 11.1%). Extended reasoning can partially compensate for the bias in some architectures, but even the best model–prompt combination does not eliminate it. Spatial position and colour pair both have negligible effect sizes, and the counterbalanced design ensures neither inflates the primary finding.

4.1 Reasoning traces

The reasoning traces from the scripted-CoT and thinking prompts illuminate how this bias may work. When asked to reason step by step, models follow a consistent pattern: describe each circle’s visual properties (always noting sharp vs. blurry edges), examine the overlap region for “uninterrupted” edges, and conclude that the sharper circle is in front. The models seem to conflate edge *sharpness* with edge *continuity*, the very feature the scripted-CoT prompt asks them to attend to. The models correctly identify the relevant visual features (one circle is crisp, the other blurred) but systematically misinterpret what those features mean for depth order.

Figure 6 presents two representative reasoning traces from bias-incongruent trials. In the first, Claude Sonnet 4.6 initially observes evidence that the blurred circle extends over the crisp one, but then explicitly reverses course and concludes the crisp circle is in front. In the second, Claude Haiku 4.5 directly maps edge sharpness to depth ordering as a general perceptual principle. Both traces illustrate the same mechanism: a verbal heuristic (“sharp means closer”) overrides the geometric occlusion information actually present in the image.

Several models go further, explicitly articulating a depth-of-field heuristic as a general perceptual principle. Claude Haiku 4.5, given

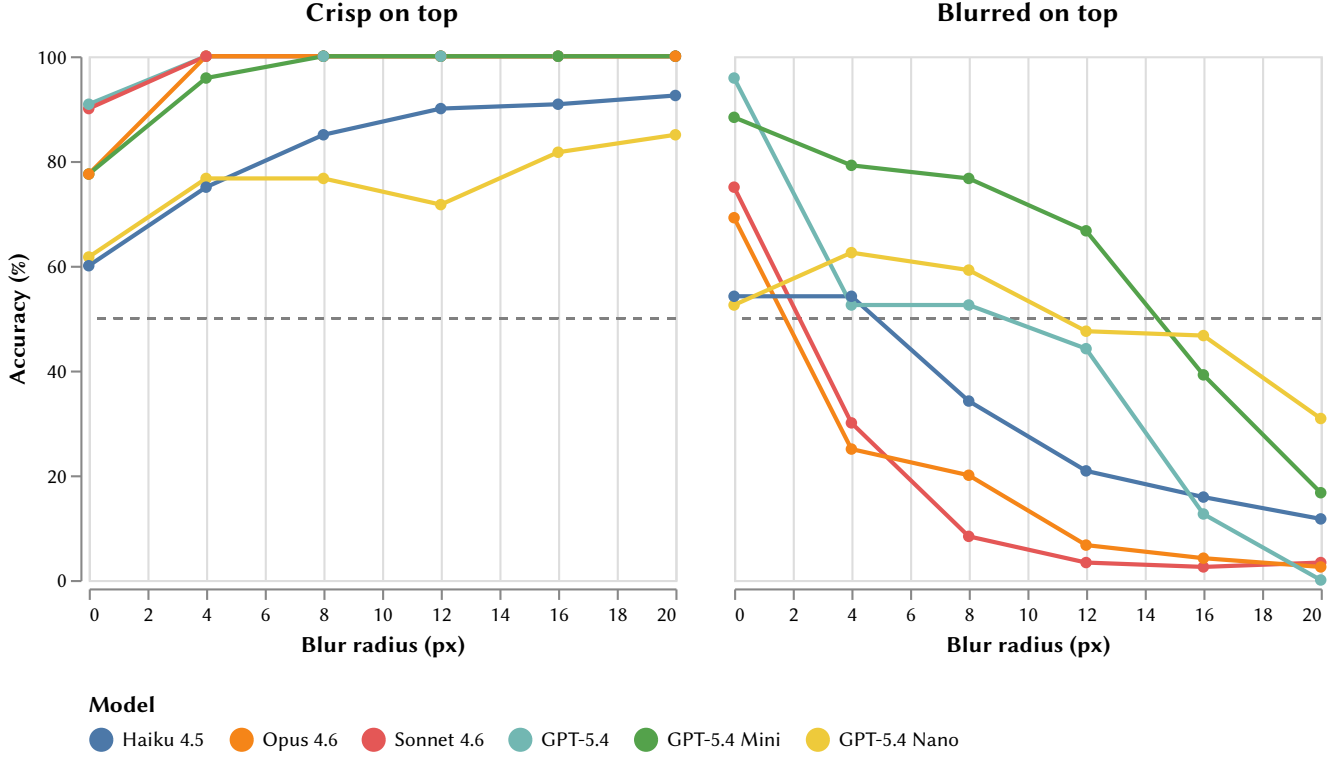


Figure 4: Accuracy by blur radius, faceted by depth order; pooled across the five non-thinking prompt variants. Left: bias-congruent trials (crisp circle on top). Right: bias-incongruent trials (blurred circle on top). Dashed line indicates 50% chance level. The thinking prompt is excluded here; its effects are reported in Table 3 and Figure 5.

the psychophysics prompt, states: “In visual perception, sharp edges typically appear closer to the viewer, while blurred edges suggest distance or being further away.” This heuristic, essentially a verbal encoding of a photography rule of thumb, is applied even though the stimuli contain no depth-of-field gradient, only occlusion edge blur. The bias operates through a verbal shortcut: the models have learned an association between sharpness and proximity that overrides the geometric occlusion information actually present in the image. Some traces show models briefly self-correcting before reverting to the crisp-in-front judgement, as in Trace A of Figure 6. The initial encoding dominates even when the reasoning process surfaces contradictory evidence—paralleling how chain-of-thought has been shown to *reinforce* rather than correct initial perceptual errors in visual illusion tasks [7].

Quantitative trace analysis. To complement these qualitative observations, we used Claude Sonnet 4.6 as an automated judge to label all 2793 reasoning traces from bias-incongruent trials along seven boolean dimensions (rubric and code in the public repository). Three patterns are robust (Table 4). First, the verbal heuristic is articulated explicitly: 27% of bias-incongruent thinking traces and 19% of scripted-CoT traces state or imply “sharp = closer”, and 13% (10% under scripted CoT) explicitly link blur to greater distance. The behaviour we infer from the answer pattern is also what the models themselves write down. Second, providers diverge sharply

on whether the trace identifies the actually-occluding (blurred) circle: GPT-5.4 thinking traces do so 66% of the time and GPT-5.4 Mini 45%, while Claude Opus, Sonnet, and Haiku do so 1–5% of the time. The OpenAI thinking models commit to their perceptual evidence; the Anthropic thinking models do not. Third, when Anthropic models do surface contradictory evidence, they tend to revert: 86% of Opus’s self-corrections (15% of all its thinking traces) ultimately conclude with the bias-congruent answer despite having noted the blurred edge extending across the overlap—the Trace-A pattern from Figure 6 writ large.

Two methodological notes. We used a single judge (Sonnet 4.6), accepting potential self-judgment bias on Anthropic-generated traces in exchange for cross-trace consistency; readers should treat per-provider differences in *articulation* labels as suggestive rather than definitive. The *identifies_correct_occluder* split is large enough to survive any plausible bias of this kind. Separately, GPT-5.4 Mini emitted no reasoning at all under the scripted-CoT prompt in 100% of bias-incongruent trials, returning only the JSON answer; the CoT instruction does not bind smaller OpenAI models in the way it does Anthropic models or larger OpenAI models.

5 Discussion

Each model we tested showed the same bias: a crisp circle is judged as being in front, regardless of whether a blurred circle is actually

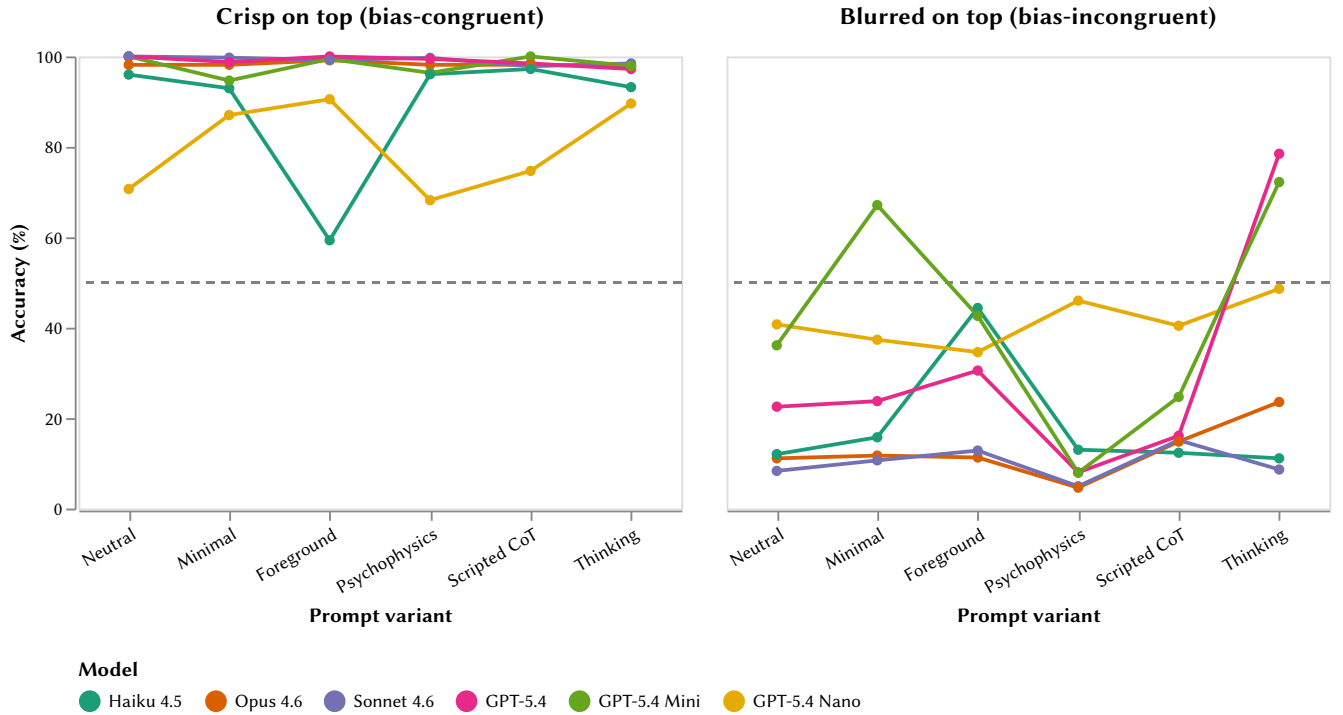


Figure 5: Effect of prompt variant on accuracy, by depth order and model. Left: bias-congruent trials (crisp on top). Right: bias-incongruent trials (blurred on top). All model-prompt combinations are near ceiling on bias-congruent trials, while most fall well below the 50% chance line (dashed) on bias-incongruent trials. Provider-level reasoning tokens (*thinking*) substantially improve GPT-5.4 and GPT-5.4 Mini but not the Anthropic models.

Trace A — Claude Sonnet 4.6 (bias-incongruent trial, blurred blue circle is on top; **incorrect answer**)

“In the overlap area, the left (golden) circle’s edge is interrupted—the right side of the golden circle fades/blends where it meets the blue circle. The right (blue) circle appears to extend behind the golden circle—but wait, let me look more carefully. Actually, the golden circle has a sharp, continuous left edge and body, but its right edge seems to fade into the blue circle. The blue circle’s left portion appears to go behind the golden circle—the blue circle’s edge on the left side is interrupted by the solid golden circle sitting in front of it.”

Trace B — Claude Haiku 4.5 (bias-incongruent trial, blurred cyan circle is on top; **incorrect answer**)

“The right circle’s edge appears sharp and well-defined, continuing clearly through the intersection area. The left circle’s edge becomes diffuse and blurred in the overlap region, appearing to fade behind the right circle. The right circle’s perimeter maintains its crisp definition even where it overlaps with the left circle. The circle whose edge continues uninterrupted through the overlap is the right circle. Its sharp, defined boundary persists through the intersection, while the left circle’s soft edge appears to be interrupted and obscured by the right circle’s presence. Therefore, the right circle is in front.”

Figure 6: Representative reasoning traces from bias-incongruent trials (blurred circle is actually in front). Trace A shows self-correction followed by reversion: the model notices the blue circle “appears to extend” over the golden one, then overrides this observation. Trace B shows direct application of the sharpness heuristic: crisp edges are equated with “continuing through” the overlap, regardless of actual depth order.

occluding it. This effect increases as the level of blur is increased and is only partially mitigated by extended reasoning tokens in some MLLM models.

Accuracy is relatively high when there is little to no blur, and falls as blur increases ruling out task confusion or random failure. But the reasoning traces complicate a simple “perception vs. reasoning” story: models *see* the blur, but draw the wrong conclusion from

it, applying a “sharp means closer” heuristic that overrides actual occlusion evidence.

The thinking-prompt results raise a sharper question: why does provider-level reasoning override the bias for OpenAI’s frontier models but barely move the Anthropic ones? The trace analysis (Table 4) suggests the gap is not about depth of reasoning but about whether the model commits to its own perceptual observations.

Table 4: Trace-judgment label frequencies for bias-incongruent thinking-prompt traces. Each cell is the per-model percentage of traces that the LLM judge (Claude Sonnet 4.6) marked TRUE for that label. Sharp⇒close: trace explicitly states or implies sharp/crisp edges indicate proximity. Blur⇒far: ditto for blur and distance. Identifies: trace at any point names the actually-in-front (blurred) circle as the occluder. Self-corr.: trace surfaces evidence contradicting its initial reading. Reverts: trace self-corrects but ultimately concludes with the bias-congruent (incorrect) answer. The provider split on *Identifies* (66%/45% for OpenAI vs 1–5% for Anthropic) is the cleanest mechanistic correlate of the answer-accuracy split in Table 3.

Provider	Model	n	Sharp⇒close	Blur⇒far	Identifies	Self-corr.	Reverts
Anthropic	Haiku 4.5	144	32.6%	24.3%	1.4%	38.2%	16.0%
Anthropic	Opus 4.6	140	56.4%	20.7%	3.6%	15.0%	12.9%
Anthropic	Sonnet 4.6	144	18.1%	3.5%	4.9%	18.8%	3.5%
OpenAI	GPT-5.4	140	19.3%	14.3%	66.4%	20.0%	1.4%
OpenAI	GPT-5.4 Mini	139	27.3%	14.4%	45.3%	17.3%	5.8%
OpenAI	GPT-5.4 Nano	143	7.0%	2.1%	2.8%	7.0%	1.4%
	Pooled	850	26.7%	13.2%	20.5%	19.4%	6.8%

GPT-5.4 thinking traces correctly identify the blurred circle as the occluder in 66% of bias-incongruent trials, while Claude Sonnet 4.6 does so under 5% of the time, despite both showing similarly high rates of attempted occlusion reasoning. The observed difference is at the conclusion stage—whether the model treats its perceptual evidence as authoritative when it conflicts with the verbal heuristic. Without access to provider training data we can only speculate about origins; candidate explanations include differences in reasoning-stage reinforcement (whether the model is rewarded for overriding initial impressions based on subsequent observations), differences in how reasoning tokens are decoded relative to answer tokens, or differences in whether perception-style tasks were included in the reasoning training mix. With only two providers tested, we also cannot rule out that the gap is specific to GPT-5.4 and Claude 4.6 rather than systematic across architectures. Future work involving open-weight models with publicly known training pipelines is the natural place to disentangle these.

This is a specific vulnerability that designers of MLLM-based forensic pipelines should be aware of. MLLMs are increasingly used in automated verification systems, assessing whether images have been manipulated and serving as backends for fact-checking tools [5, 18]. Defocus blur patterns are a physically grounded forensic signal for detecting composited media [9], and the crisp-in-front bias means that MLLMs will systematically misjudge depth relationships in images where blur is present. An adversary who understands this could exploit it, for instance, using blur to disguise depth relationships in a composited scene, essentially producing digital camouflage that evades MLLM-based detection while remaining obvious to human viewers.

This parallels a well-documented principle in biological camouflage: effective concealment does not require invisibility, only the disruption of depth and boundary cues that perceptual systems rely on to segment figure from ground [14]. In natural camouflage, edge blur and occlusion-mimicking patterns exploit precisely the low-level cues that drive depth inference, the same cues our findings suggest MLLMs weight incorrectly. The implication is that adversarially composited imagery need not be sophisticated to succeed; a manipulator who selectively degrades blur gradients at object

occlusion boundaries may produce content that is perceptually transparent to humans yet systematically opaque to automated forensic systems. This represents a qualitatively distinct threat model for MLLM-based verification pipelines, and one that existing robustness evaluations have not been designed to detect.

Concretely, consider a forensic system asked to assess whether an image has been digitally composited. Realistic image manipulation often leaves tell-tale defocus inconsistencies at the boundary of the inserted element—a known forensic signal [9]. An MLLM-based detector reasoning verbally about such an image will tend to apply the same crisp-in-front heuristic we have demonstrated: a manipulator who keeps the inserted element’s edges crisp will exploit this directly, and one who deliberately blurs the inserted element’s occluding edge will produce composites that an MLLM-based detector reads as having the manipulator’s intended depth order rather than as suspicious. The verbal-heuristic mechanism we identify in the reasoning traces (Table 4) means this failure mode is not noise but predictable from first principles: any composite where an MLLM would describe sharp/blurred edges as the principal depth cue is likely to be misjudged in the bias’s direction.

The same predictability is also a defender’s opportunity. A small calibration set of occlusion edge blur stimuli—we used 144–324 trials per model to characterise the bias, but a deployed system needs only a handful—can be inserted into a forensic pipeline’s evaluation loop as a perceptual sanity check. A system that scores below threshold on the calibration set should have its depth-related judgments flagged for human review, not because the model is broken in general but because we know exactly the class of error it will systematically make. This complements rather than replaces standard adversarial robustness testing, which is designed for pixel-level perturbations rather than the perceptual-heuristic vulnerabilities surfaced here.

More broadly, our results show the value of psychophysics-style experimental design for MLLM evaluation, as others have argued [2, 4]. Standard vision benchmarks can tell us *that* a model fails, but not *how* or *why*. Our psychophysical approach grounded in human visual perception is a useful way to begin to unpack these questions. For the disinformation detection community specifically,

this approach offers a way to identify specific perceptual vulnerabilities in MLLM components before they are exploited in the wild, rather than discovering them after the fact.

Future work should compare MLLM performance directly with human participants on the same stimuli, and investigate whether targeted training, such as reinforcement learning with verifiable rewards, can reduce the identified bias. It would also be valuable to determine whether any improvement on blurred occlusion edge perception tasks leads to positive or negative changes in model performance on other visual perception tasks.

5.1 Limitations

Our stimuli are synthetic geometric shapes (circles on a grey background), which isolate the target variable but lack the complexity of natural images. It remains to be tested whether the crisp-in-front bias transfers to naturalistic scenes with real-world textures, lighting, and multiple overlapping objects. We tested models from only two providers (Anthropic and OpenAI); other architectures, particularly open-weight models such as LLaVA or Qwen-VL, may exhibit different bias profiles. We did not collect human baseline data on our specific stimuli, relying instead on established psychophysics literature to ground our predictions about human performance. A direct human comparison would strengthen the claim that this bias is specific to MLLMs rather than a general difficulty of the task. Finally, our experiment uses a single perceptual manipulation (Gaussian blur at occlusion edges); other depth cues such as relative size, texture gradients, or aerial perspective may interact with blur in ways our design does not capture.

5.2 Reproducibility

The complete experiment is publicly available at <https://github.com/ANUCybernetics/vlm-perception-experiments>, including stimulus generation code, all generated stimuli, the evaluation harness, raw model responses (including full reasoning traces), and the analysis pipeline that produces each figure and statistic in this paper.

References

- [1] M. Asadi, J. W. O'Sullivan, F. Cao, T. Nedaee, K. Fardi, F. F. Li, E. Adeli, and E. Ashley. 2026. Mirage: The Illusion of Visual Understanding. *arXiv preprint arXiv:2603.21687* (2026).
- [2] Justin Dulay, Sonia Poltoratski, Till S. Hartmann, Samuel E. Anthony, and Walter J. Scheirer. 2024. Informing Machine Perception With Psychophysics. *Proc. IEEE* 112, 2 (2024), 88–96. doi:10.1109/JPROC.2024.3380905
- [3] Xingyu Fu, Yushi Hu, Bangzheng Li, Yu Feng, Haoyu Wang, Xudong Lin, Dan Roth, Noah A. Smith, Wei-Chiu Ma, and Ranjay Krishna. 2024. BLINK: Multimodal Large Language Models Can See but Not Perceive. In *Proceedings of the European Conference on Computer Vision (ECCV)*.
- [4] Samrajnee Ghosh, Naman Agarwal, Hemanshu Garg, Chinmay Mittal, Mausam, and Parag Singla. 2025. The Percept-V Challenge: Can Multimodal LLMs Crack Simple Perception Problems? *arXiv preprint arXiv:2508.21143* (2025).
- [5] Yiran He, Yun Cao, Bowen Yang, and Zeyu Zhang. 2025. Can GPT Tell Us Why These Images Are Synthesized? Empowering Multimodal Large Language Models for Forensics. In *Proceedings of the 2025 ACM Workshop on Information Hiding and Multimedia Security (IH&MMSec '25)*. San Jose, CA, USA. doi:10.1145/3733102.3733138
- [6] Pablo Hernández-Cámara, Alexandra Gomez-Villa, Jose Manuel Jaén-Lorites, Jorge Vila-Tomás, Valero Laparra, and Jesús Malo. 2025. Contrast Sensitivity in Multimodal Large Language Models: A Psychophysics-Inspired Evaluation. *arXiv preprint arXiv:2508.10367* (2025).
- [7] Wenjin Hou, Wei Liu, Han Hu, Xiaoxiao Sun, Serena Yeung-Levy, and Hehe Fan. 2026. Seeing Is Believing? A Benchmark for Multimodal Large Language Models on Visual Illusions and Anomalies. *arXiv preprint arXiv:2602.01816* (2026).
- [8] Anubhooti Jain, Mayank Vatsa, and Richa Singh. 2025. Words Over Pixels? Rethinking Vision in Multimodal Large Language Models. In *Proceedings of the Thirty-Fourth International Joint Conference on Artificial Intelligence (IJCAI), Survey Track*. 10481–10489.
- [9] Minsun Jeon and Simon S. Woo. 2025. Seeing Through the Blur: Unlocking Defocus Maps for Deepfake Detection. *arXiv preprint arXiv:2509.23289* (2025).
- [10] Dehong Kong, Sifan Yu, Siyuan Liang, Jiawei Liang, Jianhou Gan, Aishan Liu, and Wenqi Ren. 2025. Universal Camouflage Attack on Vision-Language Models for Autonomous Driving. *arXiv preprint arXiv:2509.20196* (2025).
- [11] Zhaochen Liu, Kaiwen Gao, Shuyi Liang, Bin Xiao, Limeng Qiao, Lin Ma, and Tingting Jiang. 2025. Beyond the visible: Benchmarking occlusion perception in multimodal large language models. *arXiv preprint arXiv:2508.04059* (2025).
- [12] John A. Marshall, Christina A. Burbeck, Dan Ariely, Jannick P. Rolland, and Kevin E. Martin. 1996. Occlusion edge blur: A cue to relative visual depth. *Journal of the Optical Society of America A* 13, 4 (1996), 681–688. doi:10.1364/JOSAA.13.000681
- [13] George Mather. 1996. Image blur as a pictorial depth cue. *Proceedings of the Royal Society B: Biological Sciences* 263, 1367 (1996), 169–172. doi:10.1098/rspb.1996.0027
- [14] S. Merilaita, N. E. Scott-Samuel, and I. C. Cuthill. 2017. How camouflage works. *Philosophical Transactions of the Royal Society B: Biological Sciences* 372, 1724 (2017).
- [15] Zenon Pylyshyn. 1999. Is vision continuous with cognition? The case for cognitive impenetrability of visual perception. *Behavioral and Brain Sciences* 22, 3 (1999), 341–365.
- [16] Pooyan Rahmzadehgervi, Logan Bolton, Mohammad Reza Taesiri, and Anh Totti Nguyen. 2024. Vision Language Models Are Blind. In *Proceedings of the Asian Conference on Computer Vision (ACCV)*.
- [17] Einat Rashal and Johan Wagemans. 2022. Depth from blur and grouping under inattention. *Attention, Perception, & Psychophysics* 84, 3 (2022), 878–898. doi:10.3758/s13414-021-02402-1
- [18] Manos Schinas and Symeon Papadopoulos. 2024. SIDBench: A Python Framework for Reliably Assessing Synthetic Image Detection Methods. In *Proceedings of the 3rd ACM International Workshop on Multimedia AI against Disinformation (MAD '24)*. Phuket, Thailand, 55–64. doi:10.1145/3643491.3660277
- [19] Zayne Sprague, Fangcong Yin, Juan Diego Rodriguez, Dongwei Jiang, Manya Wadhwa, Prasann Singhal, Xinyu Zhao, Xi Ye, Kyle Mahowald, and Greg Durrett. 2025. To CoT or Not to CoT? Chain-of-Thought Helps Mainly on Math and Symbolic Reasoning. In *Proceedings of the International Conference on Learning Representations (ICLR)*.
- [20] Gene Tangtartharakul and Katherine R. Storrs. 2026. Visual Language Models Show Widespread Visual Deficits on Neuropsychological Tests. *Nature Machine Intelligence* 8 (2026), 209–219. doi:10.1038/s42256-026-01179-y
- [21] Shengbang Tong, Zhuang Liu, Yuexiang Zhai, Yi Ma, Yann LeCun, and Saining Xie. 2024. Eyes Wide Shut? Exploring the Visual Shortcomings of Multimodal LLMs. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 9568–9578.